PAISABAZAAR MACHINE LEARNING PROJECT

Predicting Credit Risk with Machine Learning

An end-to-end data exploration, risk pattern analysis, and Random Forest classification pipeline achieving 84% accuracy.

Primary Target
Credit Score Categorization

Classifier Model
Random Forest Algorithm

Test Accuracy Score
83.96% Accuracy

PROJECT BLUEPRINT

Executive Summary & Key Takeaways

Four core insights driving credit score predictions across user accounts.

Debt Drives Risk

Unpaid debt directly inflates interest rates and lowers borrower reliability.

Income Doesn't Equal Score

High earners still default if debt management remains poor.

Delays Escalate Interest

Frequent payment delays directly trigger higher loan interest rates.

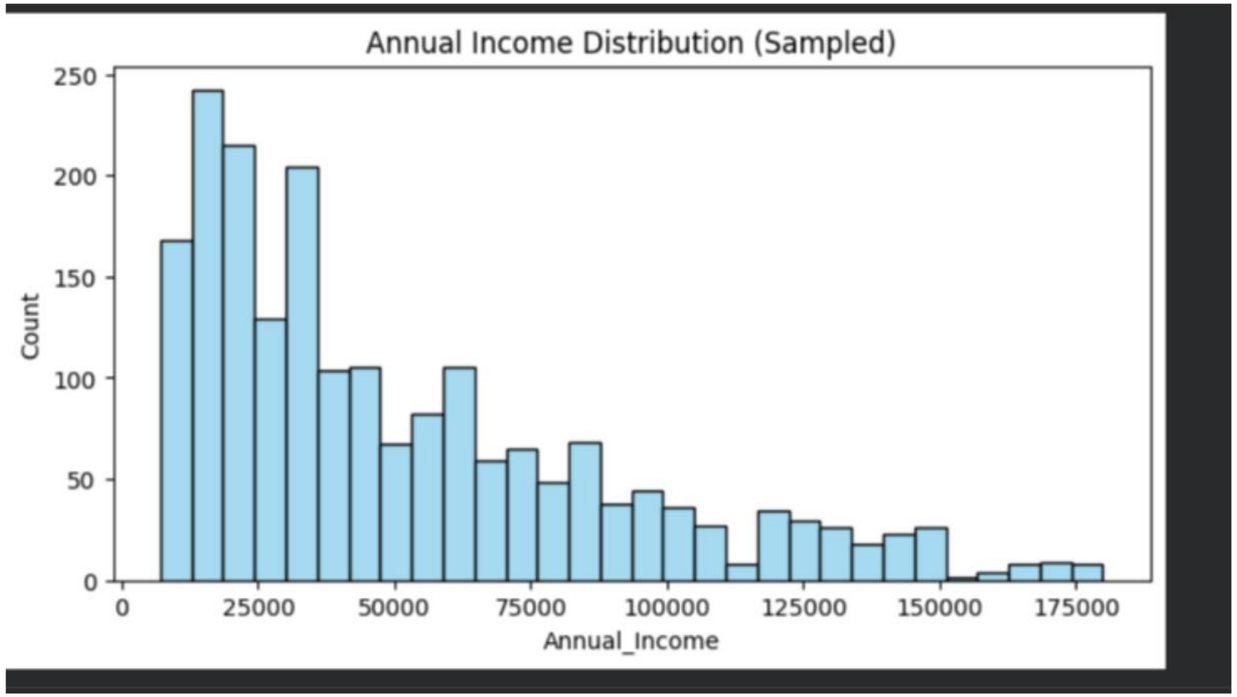
Account Clutter

Poor credit users average double the bank accounts of good users.

DEMOGRAPHIC EXPLORATION

Annual Income Distribution

Understanding the earning baseline across the borrower dataset.



Key Finding

Right-Skewed Applicant Base

The vast majority of users earn under 50,000 annually, with high earners forming a very thin tail.

Business Impact

Mass-Market Scoring

Credit scoring models must evaluate low-to-middle income behaviors with high precision.

INCOME ANALYSIS

Income vs. Monthly In-Hand Salary

Evaluating income correlation and credit score distribution.

Linear Relationship

Direct Mathematical Proportion

Monthly salary directly mirrors annual income along a straight line.

Core Takeaway

Salary Alone Doesn't Decide Score

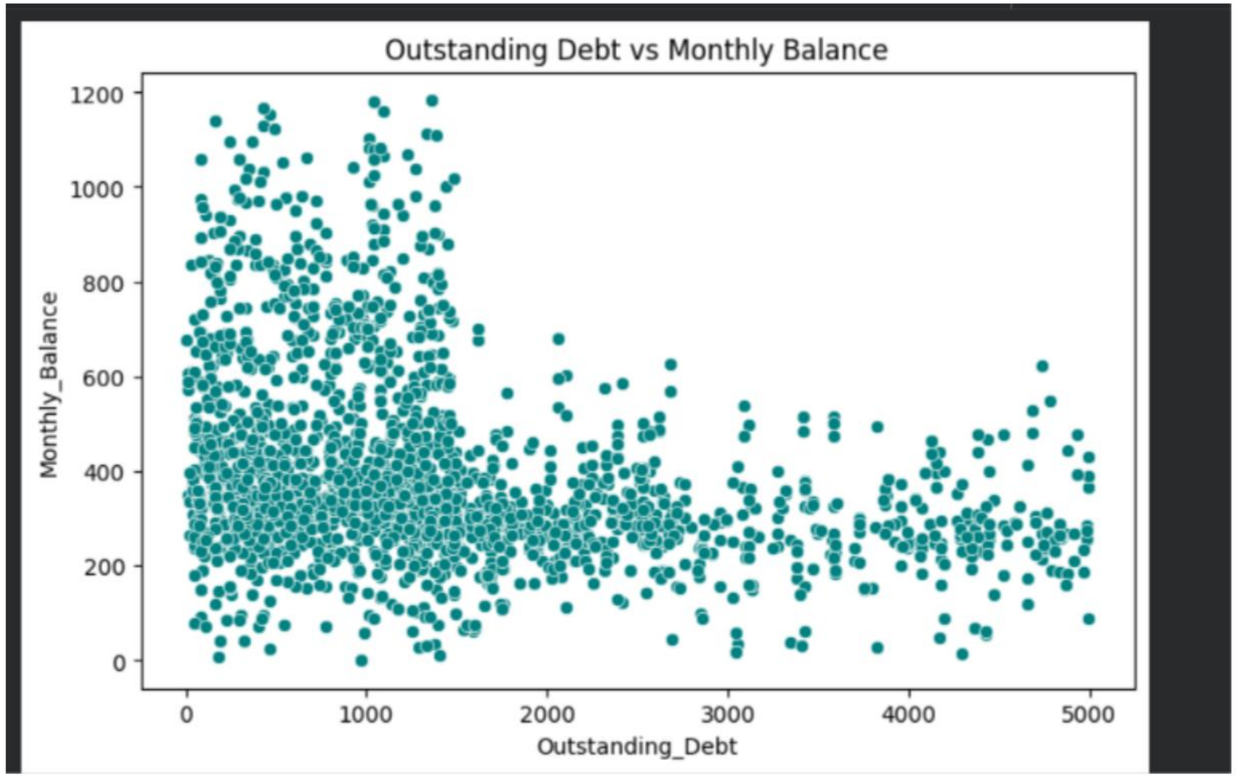
Good, Standard, and Poor credit tiers are thoroughly mixed at all salary levels.



RISK DRIVERS

Outstanding Debt vs. Monthly Balance

How heavy unpaid loans drain liquidity and elevate risk.



Low Debt Users

Healthy Cash Reserves

Borrowers with under 1,500 in debt maintain high monthly savings up to 1,200.

High Debt Users

Depleted Monthly Balance

Unpaid loans above 2,000 compress monthly balance under 400, raising default risks.

BEHAVIORAL INDICATORS

Bank Account Clutter Across Tiers

Comparing account fragmentation between credit tiers.

Good Credit Tier

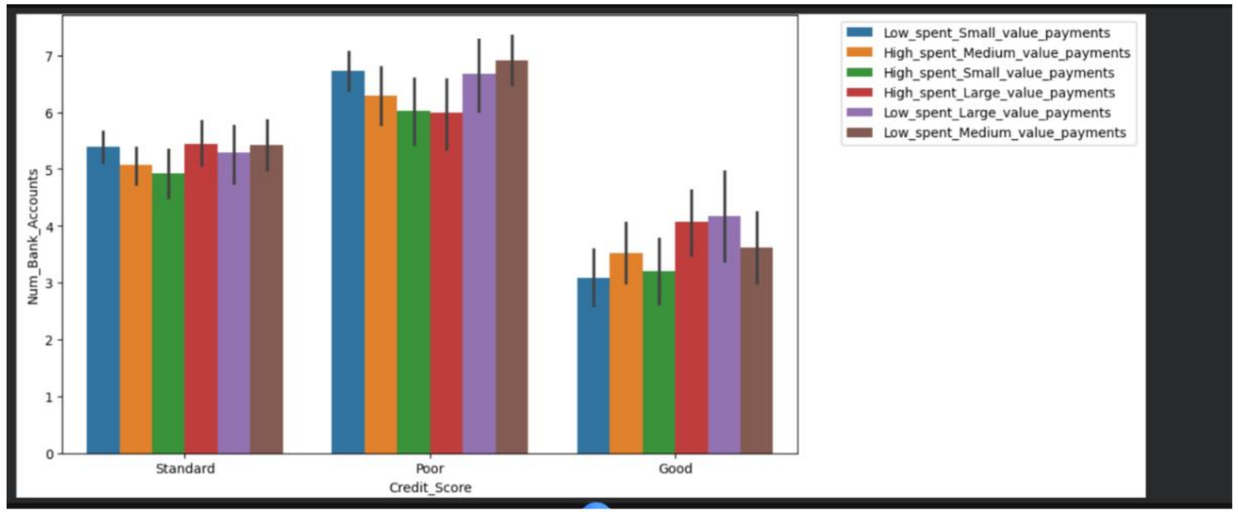
Consolidated Management

Good borrowers maintain an average of only 3 to 4 bank accounts.

Poor Credit Tier

Account Fragmentation

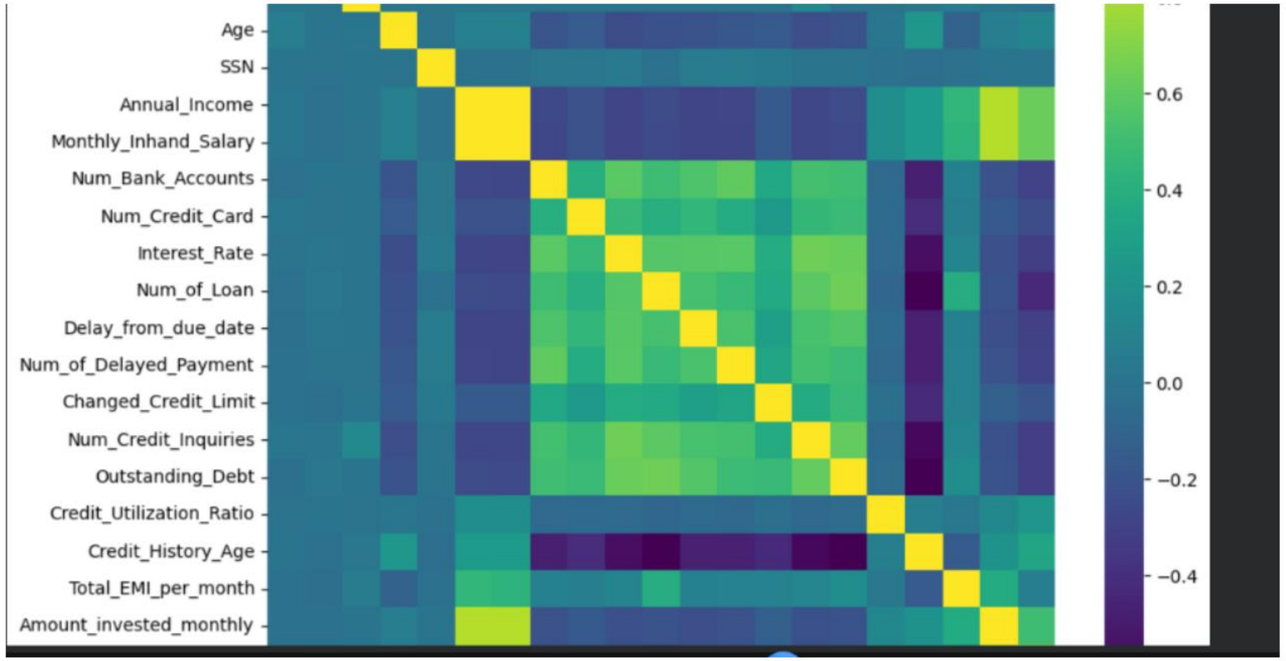
Poor borrowers hold 6 to 7 accounts, reflecting financial shuffling and risk.



MACRO CORRELATION

Full Feature Correlation Matrix

Macro-level interaction across all numerical variables in the dataset.



Positive Income Cluster

Strong yellow blocks show annual income and salary moving together perfectly.

Risk Factors Grouping

Outstanding debt, interest rates, and payment delays form a tight high-risk block.

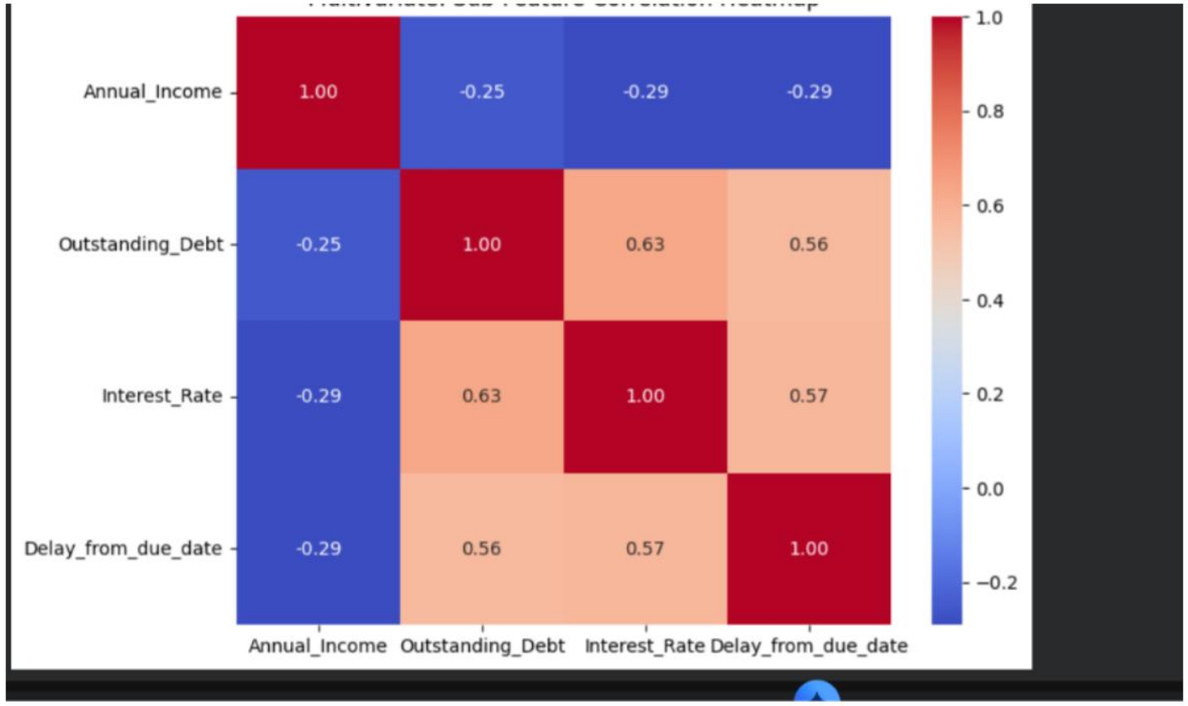
History Protective Effect

Dark purple bands show longer credit histories reducing default risk.

TARGETED FEATURE INSIGHTS

Payment Delays & Debt Escalation

Analyzing sub-feature correlations around payment delays.



Correlation: +0.56

Delay vs. Outstanding Debt

Missing payment due dates consistently builds up unpaid debt burdens.

Correlation: +0.57

Delay vs. Interest Rate

Frequent payment delays prompt lenders to charge higher interest penalties.

MACHINE LEARNING IMPLEMENTATION

Data Preprocessing & Model Setup

Preparing numeric feature matrices and splitting training data.

```
# 1. Clean missing targets & select numeric features
df_clean = df.dropna(subset=['Credit_Score']).copy()
X = df_clean.select_dtypes(include=['int64', 'float64']).drop(columns=
['Credit_Score'])
y = df_clean['Credit_Score']

# 2. Impute NaNs with median & 80/20 train-test split
X = X.apply(pd.to_numeric, errors='coerce').fillna(X.median())
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

Target Cleanup

Removes missing credit labels to ensure ground-truth training quality.

Median Imputation

Fills missing values with column medians to prevent model crash without skewing distributions.

CLASSIFICATION RESULTS

Random Forest Performance

Training ensemble trees to evaluate multi-class credit prediction accuracy.

OVERALL ACCURACY

83.96%

Random Forest Test Metric

Ensemble Voting Strength

Combines multiple decision trees to minimize overfitting and handle non-linear data boundaries.

Deployment Readiness

84% precision enables Paisabazaar partner banks to automate instant credit pre-approvals.

STRATEGIC ROADMAP

Business Recommendations

Actionable credit management strategies derived from model insights.

1

Automate Reminders

Target payment delay warnings to avoid escalating interest penalties.

2

Cap Utilization

Encourage users to maintain credit utilization below 30% thresholds.

3

Consolidate Debt

Offer debt consolidation products for high outstanding debt applicants.

PROJECT CONCLUSION

Thank You

Paisabazaar Credit Risk Classification Machine Learning Assignment

Model Performance: 83.96% Accuracy Achieved